

STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh

Program: Ph.D. in Electrical, Electronics and Computer Engineering

Proposed Advisor: Assistant Professor Van-Hai Bui

Institution: University of Michigan-Dearborn

1. Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy (PGRE) Laboratory focuses on developing distributed mathematical optimization models for decentralized energy management. While this foundation drove my graduation thesis on microgrid resilience, I recognized that classical optimization alone cannot scale to the real-time computational demands of modern power systems. I am pursuing my Ph.D. at UM-Dearborn to engineer scalable, cyber-physical control architectures that overcome these computational bottlenecks, directly advancing my long-term goal of modernizing national grid infrastructures.

2. Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering and Renewable Energy, I trained to deconstruct complex grid architectures into tractable engineering problems. Across diverse projects, I approached these challenges from multiple technical dimensions, for instance, by developing predictive models to quantify system uncertainties and prototyping power electronic circuits to evaluate hardware constraints. Maintaining this multi-faceted perspective directs my independent research trajectory and mitigates critical theoretical and physical blind spots.

My research trajectory reflects a systematic transition from treating power grids as aggregate generation-load balances to modeling them as complex computational graphs. During my early days at the PGRE lab, I formulated a Mixed-Integer Linear Programming (MILP) model for 24-hour load shifting. While this approach achieved basic economic dispatch, its reliance on a single lumped-node approximation reduced physical power flows to purely algebraic accounting. To restore operational granularity in my subsequent project, I decomposed an IEEE 123-bus network into a peer-to-peer active power trading framework governed by ADMM and Game Theory. Market transactions cleared perfectly under this distributed architecture; however, the persistent neglect of reactive power and voltage limits ultimately rendered the economic dispatch physically infeasible. Realizing that elegant algorithms are operationally invalid if the underlying grid collapses under voltage violations, I shifted toward full AC power flow modeling to bridge optimization with rigorous grid physics.

Operationalizing these AC-OPF constraints within multi-agent markets, my manuscript submitted to Sustainable Energy, Grids and Networks (SEGAN) utilized an SOCP-relaxed ADMM framework to clear P2P transactions across radial and meshed distribution networks. Extending this architecture, my graduation thesis employed Model Predictive Control (MPC) to achieve rolling-horizon benefit allocation. While both frameworks guaranteed operational safety, the profound processing overhead of continuously re-optimizing non-linear AC-OPF equations rendered classical solvers unscalable for real-time dispatch. This computational bottleneck established the technical necessity for learning-based AI control, directly motivating my decision to pursue a Ph.D.

3. Research Interests

To resolve the trade-off between computational tractability and physical grid security, my Ph.D. research will focus on architecting Physics-Informed AI frameworks for real-time power system dispatch. While pure data-driven models accelerate computational speeds, their deployment frequently fails because unconstrained black-box policies inherently violate non-linear AC power flow equations and thermal limits. I intend to synthesize these paradigms by deploying my deterministic optimization models in a dual capacity: as expert benchmarks to generate mathematically rigorous optimal dispatch datasets, and as differentiable safety

filters—leveraging convex relaxations like SOCP—that project AI-generated setpoints into a verifiable feasible region. By incorporating structural grid constraints into both the offline training and online execution agents, I aim to pioneer certifiably safe and physically resilient dispatch architectures.

4. Academic and Research Alignment

Realizing that architecting Physics-Informed AI frameworks requires an ecosystem that seamlessly bridges advanced machine learning with rigorous mathematical optimization, I found the Ph.D. program at UM-Dearborn to be the perfect catalyst for my research. To construct mathematically guaranteed safety filters for these AI agents, I require a stringent theoretical foundation. The interdisciplinary scope of the EECE curriculum will systematically bridge my theoretical gaps, providing the comprehensive mathematical and computational toolkit necessary for this architecture through advanced courses such as IMSE 505 (Optimization) and ECE 5001 (Analytic and Computational Math). Furthermore, the robust computational facilities at UM-Dearborn, paired with its strategic ties to regional energy sectors, provide the exact environment I need to rigorously test and validate these theoretical architectures against real-world grid scenarios.

Within this dynamic ecosystem, my research trajectory strongly aligns with the ongoing work of Prof. Van-Hai Bui in autonomous energy markets. His recent publications on FairMarket-RL and XRL-LLM investigate the integration of Large Language Models into multi-agent energy trading, an area highly relevant to my interest in learning-based policies. By joining his laboratory, my foundation in mathematical optimization will allow me to seamlessly support his ongoing research while advancing my own theoretical frameworks. Crucially, Prof. Bui's mentorship and profound expertise in AI surrogates will provide the exact guidance I need to successfully integrate deterministic AC safety filters into learning-based policies, ultimately realizing my vision of a physically resilient smart grid.

5. Career Objectives

Following the completion of my Ph.D, I intend to deepen my expertise as an R&D scientist within the U.S. energy sector before ultimately returning to Vietnam to contribute to the technical modernization of our national grid infrastructure. By repatriating advanced learning-based control methodologies, my long-term vision is to help transition our power architectures toward autonomous, resilient networks capable of stabilizing the rapid integration of clean energy sources. Ultimately, the Ph.D. program at the University of Michigan-Dearborn provides the exact interdisciplinary ecosystem and mentorship necessary to launch this career trajectory.